



Unix Command Line Interface (CLI)

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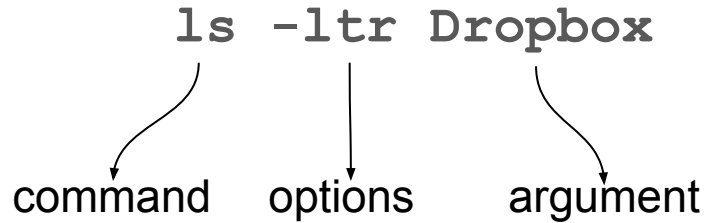


Unix Commands

- Unix has a set of predefined commands that help a user in interacting with the OS:
 - Manage files and directories.
 - Login to a remote system
 - Identify processes taking large CPU time, kill a hung process.
 - Printing a document.
- Variations of the command are introduced by switches or **options**.
- Commands can be combined by using operators.
- The interaction can take two forms:
 - **Command line Interface (CLI)**: Interaction with Unix from the command line in a terminal.
 - **Bash script**: Interaction by placing commands in a file.



Example of a command



Meaning: List the contents of the directory Dropbox

- In the long format
- sorted by time
- in reverse order (latest at the end)



General syntax of a command

> `command option(s) argument(s)`

- In the command `ls -ltr Dropbox`, each of `l`, `t` and `r` is an option
- Some (but not all) options can be written in a verbose fashion:

```
ls -lt --reverse Dropbox
```

```
ls -l -t --reverse Dropbox
```

- The command to find information about commands is also a command called **man**

man ls: Gives information about `ls`



File management commands

cat	Concatenate files and displays the result
cp, mv	Copy, move (rename) files (or directories)
ls, tree	List files and directories.
chmod, chgrp, chown	Change access modes, group or owner on files.
mkdir	Create a directory.
ln, ln -s	Create a hard, symbolic link.
pwd	Print working directory
rm, rmdir	Remove files, directories
less	Display files by screenful, forward and backward
wc	Count lines, words, and characters.
head, tail	Show the first few lines (last few lines) of a file.



ls -l

```
sanyal@sanyal:~/Dropbox/cs251/bash$ ls -al
total 120
```

permission	no of links	owner	group	size	date	name
drwxrwxr-x	4	sanyal	sanyal	4096	Aug 3 21:16	.
drwxrwxr-x	15	sanyal	sanyal	4096	Jul 31 11:31	..
-rw-rw-r--	1	sanyal	sanyal	0	Aug 3 21:01	.a.txt
-rw-rw-r--	1	sanyal	sanyal	0	Aug 2 18:53	bar.txt
-rw-rw-r--	1	sanyal	sanyal	399	Aug 3 21:16	bash-examples.sh
-rw-rw-r--	1	sanyal	sanyal	262	Aug 3 10:25	bash-examples.sh~
-rw-rw-r--	1	sanyal	sanyal	87349	Aug 1 12:14	bash-scribbles.xoj
-rw-rw-r--	1	sanyal	sanyal	0	Aug 2 18:54	foo.txt
-rw-----	1	sanyal	sanyal	136	Aug 3 21:04	history
drwxrwxr-x	3	sanyal	sanyal	4096	Aug 2 19:27	tmp
drwxrwxr-x	3	sanyal	sanyal	4096	Aug 2 19:40	tmp1



Owner and group

- Owner and group
 - Each user is assigned a primary group
 - The system administrator can introduce a new group and populate the group with users
 - When a user creates a file he is the default owner and his primary group is the default group
 - The group can be changed with `chgrp`
 - Every other user in the same group has permission defined by the second triad.



permission

Three permission triads

first triad	what the owner can do
second triad	what the group members can do
third triad	what other users can do

Each triad

first character	r : readable
second character	w : writable
third character	x : executable

```
-rwx--x--- 1 sanyal testgroup 136 Aug 3 21:04 test.sh
drwxrw-r-x 3 sanyal cs251 4096 Aug 2 19:27 tmp
```




permission

```
-rwx--x--- 1 sanyal testgroup 136 Aug 3 21:04 test.sh
```

- Not a directory (-)
- owner has read, write and execute permissions
- group member has execute permission
- others have no permission

```
drwxrw-r-x 3 sanyal cs251 4096 Aug 2 19:27 tmp
```

- Is a directory (d)
- owner can list files (r), change the directory contents (wx) and step into (cd) the directory (x).
- group members can just list files(r). The w bit is meaningless if x is not set.
- others can only list files (r) and step into the directory. They can change files but not add or delete files in the directory



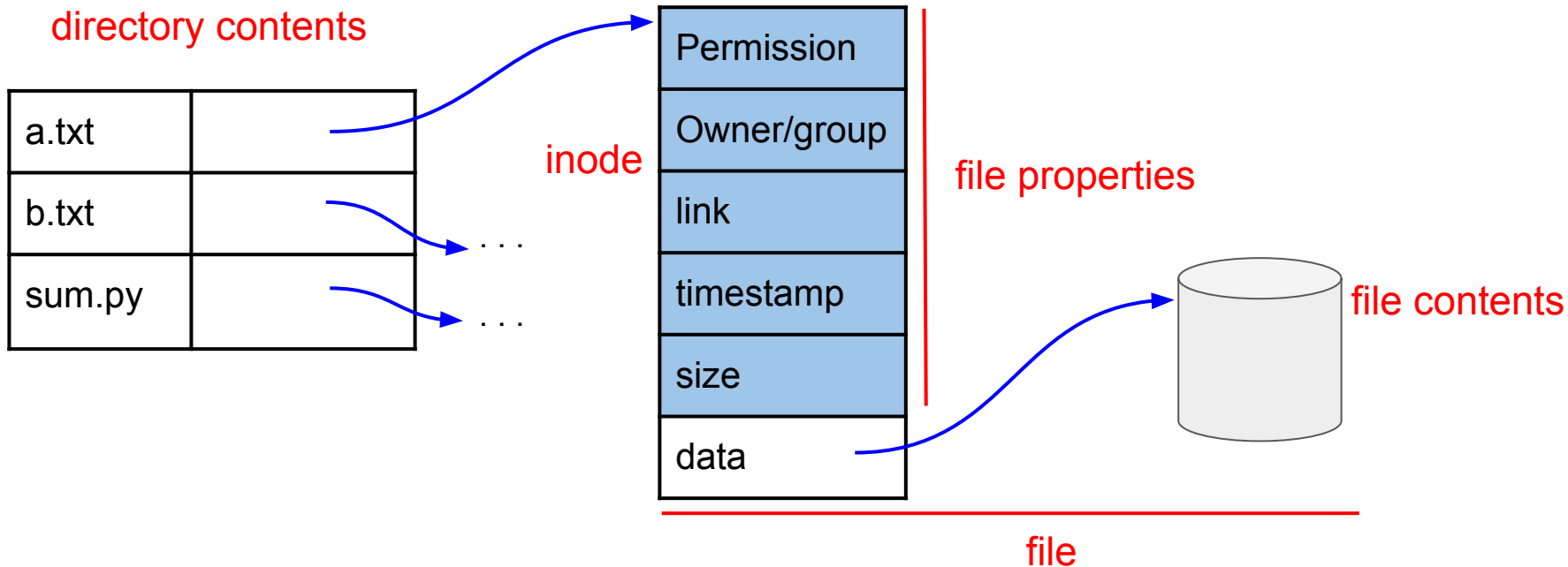
changing permission with chmod

chmod NNNN filename

Symbolic Notation	Numeric Notation	English
-----	0000	no permissions
-rwx-----	0700	read, write, & execute only for owner
-rwxrwx---	0770	read, write, & execute for owner and group
-rwxrwxrwx	0777	read, write, & execute for owner, group and others
---x--x--x	0111	execute
--w--w--w-	0222	write
--wx-wx-wx	0333	write & execute
-r--r--r--	0444	read
-r-xr-xr-x	0555	read & execute
-rw-rw-rw-	0666	read & write
-rwxr-----	0740	owner can read, write, & execute; group can only read; others have no permissions

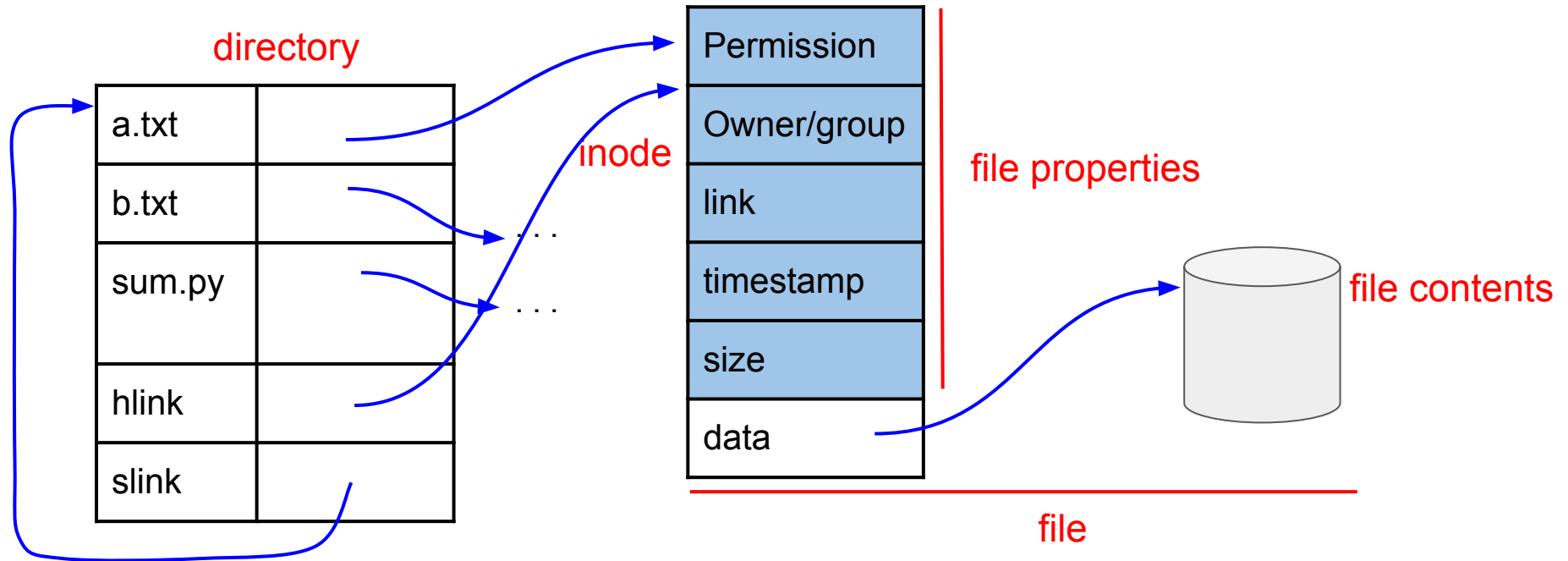


File System Representation





Hard links, soft links and the `ln` command



- `ln a.txt hlink` -- creates the hard link `hlink`
- `ln -s a.txt slink` creates the symbolic link `slink`
- Links are normally used to call a new version of a program with an old name.



File comparison and searching

cmp	Compare two files, byte by byte.
comm	Compare items in two sorted files.
diff , meld , diff3	Compare two (three) files, line by line.
find	Search the system for files by name and take a range of possible actions.
grep	Search files for text patterns.
which	Print pathname of a command.
whereis	Find command.
locate	Search a pre-existing database to show where files are on the system.



Comparing files with `diff`

- `diff file1.txt file2.txt`: Compares files `file1.txt` and `file2.txt`. Indicates changes that have to be made to `file1.txt` to make it identical to `file2.txt`.
- Example:

```
This is a first line
This is a second line
This is a third line.
This is the last line.
And that is all
```

```
This is the zeroth line.
This is a first line
This is a second line.
    This is a third line
This is the last line.
```

```
0a1
> This is the zeroeth line.
2,3c3,4
< This is a second line
< This is a third line.
---
> This is a second line.
> This is a third line
5d5
< And that is all
```



Comparing files with `diff`

file1.txt:

```
This is a first line
This is a second line
This is a third line.
This is the last line.
And that is all
```

file2.txt:

```
This is the zeroth line.
This is a first line
This is a second line.
    This is a third line
This is the last line.
```

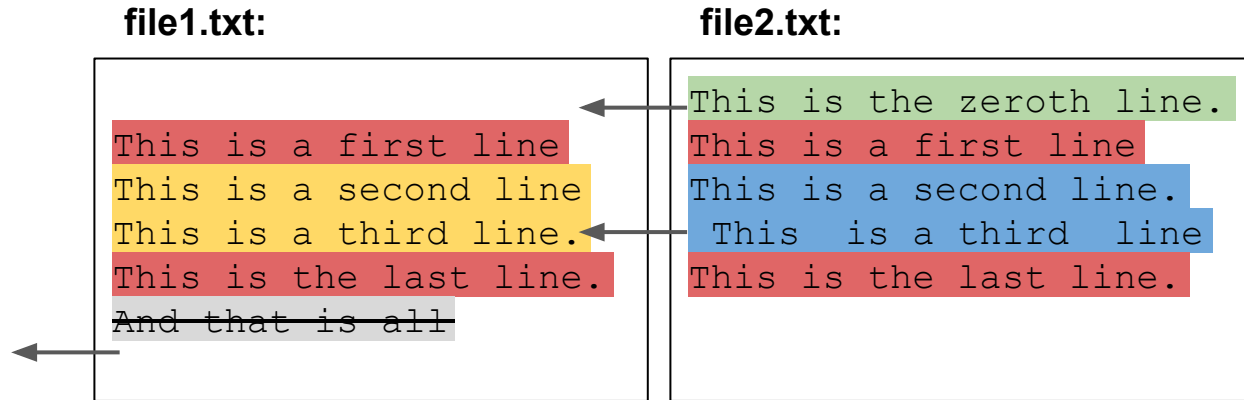
- `0a1`: add at position 0 of the first file the line “**This is the zeroeth line**” to match line 1 of the second file.
- `2,3c3,4`: change lines 2,3 as shown below to match lines 3,4 of the second file.
 `< This is a second line`
 `< This is a third line.`
 `---`
 `> This is a second line.`
 `> This is a third line`
- `5d5`: delete line 5 (shown below) of the first file to match line 5 of the second file.
 `< And that is all`

```
diff file1.txt file2.txt
```

```
0a1
> This is the zeroeth line.
2,3c3,4
< This is a second line
< This is a third line.
---
> This is a second line.
> This is a third line
5d5
< And that is all
```



Comparing files with `diff`



All changes are made to the first file.



Searching for patterns with `egrep` or `grep -E`

- `egrep`: Extented global regular expression print
- `egrep <pattern> <filename>`: Prints every line in `<filename>` containing the pattern described by `print`.
- Examples:

<code>egrep line grep.txt</code>	Print all lines containing the pattern <code>line</code>
<code>egrep 'a f' grep.txt</code>	Print all lines containing <code>a f</code> .
<code>egrep 'first second' grep.txt</code>	Print all lines containing <code>first</code> or <code>second</code>
<code>egrep '(Th)?is' grep.txt</code>	Print all lines containing <code>This</code> or <code>is</code>
<code>egrep '[a-z]{2,5}' grep.txt</code>	...strings made of characters <code>a to z</code> and length 2,3,4 or 5.
<code>egrep '[a-z]{5}' grep.txt</code>	...strings made of characters <code>a to z</code> and length 5.



Searching for patterns with `egrep` or `grep -E`

<code>egrep '19(D 0)050[0-9]{3}' Students.csv</code>	Students who had originally joined CSE
<code>egrep -v '19(D 0)050[0-9]{3}' Students.csv</code>	Others
<code>egrep '\"S' Students.csv</code>	How many names start with S?
<code>egrep '\"([[:alpha:]]+){m}([[:alpha:]]+)\"'</code> <code>Students.csv</code>	How many names have <code>m+1</code> words in them? Notice the space after the +
<code>ls -al egrep '^d'</code>	Which files are directories
<code>egrep</code> <code>'^([a-zA-Z0-9_\\-\\.]+)@([a-zA-Z0-9_\\-\\.]+)\\.([a-zA-Z]{2,5})\$'</code> <code>emails</code>	Lines starting with valid email addresses



Difference between egrep and grep

- grep recognizes uses Basic Regular Expressions (BRE) which does not include the characters {, }, (,), | and + as special characters. They have to be “escaped” using \.

```
> egrep 'a|b' grep.txt
```

```
This is a first line  
This is a second line  
This is a third line.  
This is the last line.  
And that is all  
Funny line a|b
```

```
> grep 'a|b' grep.txt
```

```
Funny line a|b
```



System status

date	Display or set date.
df	Show free disk space.
du	Show disk usage.
env	Environment variables.
ps	Show processes.
stat	Display file or filesystem status.
uname	Display system information
kill, pkill	Terminate a running process.



Disk partitions

- **Disk partitioning** is the creation of one or more regions, so that each region can be managed separately.^[2] These regions are called partitions. On my disk: `/dev/nvme0n1p1`, `/dev/nvme0n1p7` and `/dev/nvme0n1p9`.
- Different partitions of the disk can be mounted on directories of the Unix mount tree. On my system:
 - `/dev/nvme0n1p7` is mounted on `/`
 - `/dev/nvme0n1p9` is mounted on `/home`
 - `/dev/nvme0n1p1` is mounted on `/boot/efi`
- Partitions and their mount points can be found using the **df** command
- Separating user data from system data can prevent the system partition from becoming full and rendering the system unusable.



Text processing and others

cut	Extracts fields of a line separated by delimiters.
fmt	Convert text to specified width by filling lines and removing newlines.
paste	Merge corresponding lines of one or more files into tab-separated vertical columns and write to standard output.
sort	Sort lines by key
tr	Translate characters.
expand	Convert tabs in given files to an appropriate number of spaces
wget	Access files through the Internet
basename	Remove leading directory components from a path.



Resources

[Cheat sheet](#) Keep it by your side.

[Advanced bash scripting guide](#) Note, “advanced”.

[All about regular expressions](#) Bookmark this.

[The best that one can find about **find**](#)

[The **tr** command](#)

[Disk partitioning](#)

[Monitoring processes with **ps**](#)

[The **basename** command](#)

[**wget** it?](#)